

Python Programming Assignment

1: Write a Python program using `multiprocessing.Pool` to calculate the sum of all even numbers from 1 to N for every number from the given list.

Input

```
Data = [1000000, 2000000, 3000000, 4000000]
```

Expected Task

For each number N, calculate:

$$2 + 4 + 6 + \dots + N$$

Expected Output Format

```
Process ID : 1234  
Input Number : 1000000  
Sum of Even Numbers : 250000500000
```

2: Write a Python program using `multiprocessing.Pool` to calculate the sum of all odd numbers from 1 to N.

Input

```
Data = [1000000, 2000000, 3000000, 4000000]
```

Expected Task

For each number N, calculate:

$$1 + 3 + 5 + \dots + N$$

Expected Output Format

```
Process ID : 1235  
Input Number : 1000000  
Sum of Odd Numbers : 250000000000
```

3: Write a program that counts how many even numbers exist between 1 and n using `Pool.map()`.

Input

```
Data = [1000000, 2000000, 3000000, 4000000]
```

Expected Output Format

```
Process ID : 1236  
Input Number : 1000000  
Even Number Count : 500000
```

4: Write a program that counts how many odd numbers exist between 1 and n .

Input

```
Data = [1000000, 2000000, 3000000, 4000000]
```

Expected Output Format

```
Process ID : 1237  
Input Number : 1000000  
Odd Number Count : 500000
```

5: Write a program that calculates factorials of multiple numbers simultaneously using `multiprocessing.Pool`.

Input

```
Data = [10, 15, 20, 25]
```

Expected Task

For every N , calculate:

$N!$

Expected Output Format

```
Process ID : 1240  
Input Number : 20  
Factorial : 2432902008176640000
```